

Bovnar Python Bindings

Bovnar (BVNR) v1.1 documentation · Python Bindings · 2026-06-16

Version: 1.1 (draft)

Pure-`ctypes` Python bindings for the **Bovnar (BVNR)** typed serialisation library (spec v1.1, draft).

No compiled extension module is needed — the bindings load `libbvnr_shared.so` at import time via the standard `ctypes.CDLL` machinery.

Requirements

Requirement	Notes
Python $\geq$ 3.10	<code>dataclasses</code> , <code>enum.IntEnum</code> , union-type annotations (<code>X Y</code>)
<code>libbvnr_shared.so</code>	Runtime only; see Library discovery below
<code>numpy</code> $\geq$ 1.24	Optional — only for the NumPy bridge (<code>pip install bovnar[numpy]</code>)
<code>pint</code> $\geq$ 0.22	Optional — only for the pint bridge (<code>pip install bovnar[pint]</code>)
<code>pytest</code> $\geq$ 7	Test suite only (<code>pip install bovnar[dev]</code>)

`numpy` and `pint` are imported **lazily, on first use** of their respective bridge functions — importing `bovnar` never requires either to be installed.

Installation

```
# Editable install from source (recommended during development)
pip install -e ".[dev]"
```

Optional extras pull in the dependencies for the bridges:

```
pip install "bovnar[numpy]" # NumPy bridge
pip install "bovnar[pint]"  # pint bridge
pip install "bovnar[all]"   # both numpy and pint
```

Library discovery

The bindings search for `libbvnr_shared.so` in this order:

1. `LIBBOVNR_PATH` — absolute path to the `.so` file, e.g. `bash export LIBBOVNR_PATH=/opt/bovnr/lib/libbvnr_shared.so`
2. `LIBBOVNR_DIR` — directory that contains the `.so`, e.g. `bash export LIBBOVNR_DIR=/opt/bovnr/lib`
3. `ctypes.util.find_library('bovnr')` — standard `ldconfig` / `LD_LIBRARY_PATH` search.
4. **In-tree build paths** — `../../build/`, `../../build/release/`, `../../`, `.` (resolved relative to the `_ffi.py` file; useful when building Bovnar alongside the bindings from a mono-repo).

If the library cannot be found a `BovnarLibraryNotFound` exception is raised with the list of searched paths.

Quick-start

High-level API (`loads` / `dumps`)

```
import bovnr

# Serialize a Python dict to BVNR bytes
data = {
    "sensor_id": 42,
    "temperature": -3.7,
    "label": "outdoor",
    "active": True,
    "payload": None,
}
bvnr_bytes = bovnr.dumps(data)

# Deserialise back to a Python dict
recovered = bovnr.loads(bvnr_bytes)
assert recovered["sensor_id"] == 42
```

`loads` accepts an optional `typed` flag that wraps typed values in `Quantity` objects instead of decoding them to native Python scalars. This preserves the original text representation, exact type width, numeral base, and physical unit for lossless round-trips:

```
bvnr = b".pressure = <float:32,Pa> 101325.0;"
doc = bovnr.loads(bvnr, typed=True)
q = doc["pressure"] # Quantity('101325.0', FLOAT [Pa])
print(q.raw)        # '101325.0'
print(q.unit_str())  # 'Pa'

# dumps() accepts Quantity values – annotation and raw text are re-emitted as-is
out = bovnr.dumps(doc)
assert bovnr.loads(out, typed=True) == doc
```

`dumps()` starts with a 4 MiB write buffer and doubles it automatically on overflow, up to 256 MiB. The `cap` keyword argument that existed in earlier versions is no longer accepted.

SAX-style streaming reader

The verified callback receives exactly **two** positional arguments: the event code and the data payload. There is no userdata/context argument — capture external state via closure instead.

```
from bovnar.reader import Reader
from bovnar import Event

def on_event(ev, data):
    if ev == Event.ASSIGNMENT_START:
        print("key:", data.raw_str())
    elif ev == Event.DATA:
        print("value bytes:", data.raw_bytes())
    return True # returning False stops parsing (raises BovnarParseError)

with Reader() as r:
    r.read_mem(bvnr_bytes, on_verified=on_event)
```

Generator / iterator interface

```
from bovnar.reader import Reader

with Reader() as r:
    for payload in r.iter_mem(bvnr_bytes):
        print(payload.event, payload.text)
```

DOM (random-access) API

```
import bovnar

doc = bovnar.dom_parse(bvnr_bytes)

# Top-level key lookup
node = doc["sensor_id"]
print(node.as_i64()) # → 42
print(node.value_type) # ValueTypeSpec(family=UINT, width=64, base=0)

# Struct traversal
cfg = doc["config"]
host = cfg["host"].as_str()

# Array access
arr = doc["values"]
for i in range(len(arr)):
    print(arr[i].as_float())

# Convert entire document to a plain Python dict (drops type/unit info)
d = doc.to_dict()
```

Low-level writer

```
from bovnar.writer import Writer
from bovnar.enums import BaseUnit, SIPrefix

with Writer.to_mem() as w:
    w.write_float("velocity", 9.81, width=64,
                  unit_si_base=BaseUnit.METER,
                  unit_si_prefix=SIPrefix.NONE)
    w.write_uint("count", 1024, width=32)

output: bytes = w.get_output()
```

Streaming / framing (bovnar.stream)

Bindings for the streaming layer (see [Streaming, Framing & Multiplexing](#) for the full treatment):

```
from bovnar import stream

# Multi-document record framing
blob = stream.dump_documents([{"id": 1}, {"id": 2}]) # list[dict] -> bytes
docs = stream.load_documents(blob) # bytes -> list[dict]
docs = stream.load_documents(blob, continue_past_failed=True) # bad docs -> None

# Octet multiplexing: (channel, payload) of any size, interleaved & reassembled
multiplexed = stream.mux_dump([(1, b"hello"), (42, b"world")])
messages = stream.mux_load(multiplexed) # [(1, b"hello"), (42, b"world")]

# Document-in-document
outer = stream.embed_document(bovnar.dumps({"v": 1}), key="payload")
inner = stream.parse_embedded(bovnar.loads(outer)["payload"]) # {"v": 1}
```

Unit helpers

```
import bovnar

# Parse a unit string into a ValueUnit struct
vu = bovnar.parse_unit("k~g·m/s²")

# Convert a ValueUnit back to its canonical string
s = bovnar.unit_to_str(vu) # → "k~g·m/s²"

# Scalar SI/IEC prefix factor for a unit string
f = bovnar.unit_factor("M~Hz") # → 1_000_000.0
```

Extended unit functions

The following functions operate on `ValueUnit` objects and are available both from the top-level `bovnar` namespace and from `bovnar.units`.

```

from bovnar import (
    unit_valid, unit_prefix_factor, unit_prefix_exponent,
    prefix_unit_valid,
    unit_to_si_factor, units_compatible,
    unit_convert_factor, unit_dimension_vector,
    unit_reduce, unit_to_str_ex,
    exponent_to_int, int_to_exponent,
    convert_value,
    UnitFlags,
)

# Validate a ValueUnit struct
ok = unit_valid(vu) # True when vu is structurally valid

# Prefix scale factor (SI or IEC) for a ValueUnit
f = unit_prefix_factor(vu) # e.g. 1000.0 for k~m, 2**30 for Gi~B

# Prefix exponent (base-10 for SI, base-2 for IEC)
e = unit_prefix_exponent(vu) # e.g. 3 for kilo, -3 for milli, 30 for gibi

# Validate a prefix for a base unit (IEC prefixes are only valid on bit/byte)
from bovnar import ValueUnitPrefix, IECPrefix, BaseUnit
p = ValueUnitPrefix.make_iec(IECPrefix.GIBI)
ok = prefix_unit_valid(p, BaseUnit.BYTE) # True
ok = prefix_unit_valid(p, BaseUnit.METER) # False

# Full SI conversion including affine terms (e.g. Celsius → Kelvin)
conv = unit_to_si_factor(vu)
# conv.factor, conv.is_affine, conv.affine_offset

# Check dimensional compatibility
ok = units_compatible(vu_a, vu_b) # True if same SI dimension vector

# Conversion factor between two compatible units
c = unit_convert_factor(vu_from, vu_to)
# c.factor, c.requires_affine

# 7-element SI dimension exponent vector [m, kg, s, A, K, mol, cd]
dims = unit_dimension_vector(vu) # e.g. [1, 0, -1, 0, 0, 0, 0] for m/s

# Reduce a compound unit to its canonical named SI unit
r = unit_reduce(vu) # r.unit, r.scale

# Convert a scalar value between units (handles affine conversions)
kelvin = convert_value(25.0, vu_celsius, vu_kelvin)

# Serialise with formatting options (see UnitFlags below)
s = unit_to_str_ex(vu, UnitFlags.ASCII_EXP) # use ^N instead of Unicode superscripts
s = unit_to_str_ex(vu, UnitFlags.REDUCE) # reduce to canonical named unit first
s = unit_to_str_ex(vu, UnitFlags.REDUCE | UnitFlags.ASCII_EXP)

# Exponent enum ↔ integer conversions
n = exponent_to_int(Exponent.NEG_SQUARE) # → -2
exp = int_to_exponent(-2) # → Exponent.NEG_SQUARE

```

`SI_DIM_NAMES` is the ordered tuple `('m', 'kg', 's', 'A', 'K', 'mol', 'cd')` — the index positions used by `unit_dimension_vector`.

UnitFlags

```
from bovnar import UnitFlags  # also from bovnar.units

UnitFlags.NONE      # 0 – no special formatting
UnitFlags.REDUCE     # reduce to a canonical named SI unit before serialising
UnitFlags.ASCII_EXP  # use ^N exponent notation instead of Unicode superscripts
```

`UnitFlags` is an `IntFlag` and its values may be OR-combined:

```
s = unit_to_str_ex(vu, UnitFlags.REDUCE | UnitFlags.ASCII_EXP)
```

ValueUnitPrefix

`ValueUnitPrefix` is the public mirror of the C `value_unit_prefix_t` struct. It can be constructed with class methods or extracted from a `ValueUnitComponent`:

```
from bovnar import ValueUnitPrefix, SIPrefix, IECPrefix

p_si = ValueUnitPrefix.make_si(SIPrefix.KILO)
p_iec = ValueUnitPrefix.make_iec(IECPrefix.GIBI)

vu = bovnar.parse_unit("Gi~B")
comp = vu.components[0]
p = comp.prefix  # ValueUnitPrefix extracted from a component
```

Inline unit suffix

In addition to the unit embedded in a type annotation (`<float:64,m/s>`), the Bovnar format supports an **inline unit suffix** placed directly after a scalar value, before the terminating `;`:

```
.speed = 9.81 m/s;          # inline suffix, no annotation
.mass  = <float:64> 70.5 k~g; # annotation without unit, inline adopted
.dist  = <float:64,m> 1.5 m; # annotation and inline both say 'm' – valid
```

From the Python layer, inline and annotation units are **identical**: both reach the application as `data.value_unit` inside the `Event.DATA` payload. No extra code is required to consume inline units.

The validator raises `ErrorCode.UNIT_MISMATCH` (38 / `BovnarParseError`) when an inline suffix is present and a type-annotation unit is also present but the two do not resolve to the same `value_unit_t`. Inline unit suffixes inside array elements always raise `ErrorCode.UNEXPECTED_INPUT_BYTE`.

Quantity

`Quantity` is a typed, unit-annotated scalar value that preserves the original text representation, type width, numeral base, and physical unit across a `loads` / `dumps` round-trip.

```
from bovnar import Quantity, ValueTypeSpec, ValueUnit
from bovnar.enums import ValueTypeFamily
```

Construction

`Quantity` is normally produced by `loads(..., typed=True)` rather than constructed by hand, but direct construction is supported:

```
from bovnar.structs import make_type_spec
from bovnar.enums import ValueTypeFamily

vt = make_type_spec(ValueTypeFamily.FLOAT, 32)
q = Quantity('101325.0', vt)          # dimensionless
q2 = Quantity('9.81', vt, vu)         # vu is a ValueUnit, e.g. from parse_unit
```

Properties and methods

Name	Type	Description
<code>q.raw</code>	<code>str</code> <code>None</code>	Original text token as it appeared in the BVNR stream
<code>q.vtype</code>	<code>ValueTypeSpec</code>	Type family, bit width, and numeral base
<code>q.unit</code>	<code>ValueUnit</code>	Physical unit (<code>BVN_UNIT_NONE</code> when dimensionless)
<code>q.value</code>	property	Decode <code>raw</code> to the closest native Python scalar (<code>int</code> , <code>float</code> , <code>str</code> , <code>bool</code>)
<code>q.unit_str()</code>	<code>str</code>	Canonical unit string (e.g. <code>'m/s²'</code>), or <code>''</code> when dimensionless

`dumps()` integration

`_emit_value` dispatches on `Quantity` before the plain `int` / `float` path, so any dict that came from `loads(..., typed=True)` can be passed directly to `dumps()` and will produce identical output:

```
bvnr = b".speed = <float:32,m/s> 9.81;"
doc = bovnar.loads(bvnr, typed=True)
out = bovnar.dumps(doc)
assert out.strip() == bvnr.strip()
```

The annotation is suppressed when `_needs_annotation` returns `False` — i.e. when the type is `FLOAT:64` with no unit and no non-decimal base (matching the C library's own default-annotation omission rules).

`write_array` helper

`write_array` is a high-level helper exported from the top-level `bovnar` namespace. It handles both flat and multi-dimensional arrays.

```
from bovnar import write_array, Writer
from bovnar.structs import make_type_spec, make_unit_si
from bovnar.enums import ValueTypeFamily, BaseUnit

with Writer.to_mem() as w:
    # Single-row array
    write_array(w, "primes", [2, 3, 5, 7])

    # Multi-row array (rows separated by /)
    write_array(w, "matrix", [[1, 2, 3], [4, 5, 6]])

    # Typed array: whole-array type annotation
    vt = make_type_spec(ValueTypeFamily.UINT, 16)
    write_array(w, "ports", [80, 443, 8080], vt=vt)
```

Element types accepted per element: `int`, `float`, `str`, `bool`, `None`, `dict` (written as a struct), or nested `list` / `tuple` (written as a nested array). When rows is a flat sequence it is treated as a single-row array; when all top-level elements are themselves `list` or `tuple` it is treated as a multi-row array.

pint bridge

bovnar units interoperate with `pint` through a hand-verified translation table (`bovnar._pint_units`). `pint` is an **optional** dependency, imported lazily on first use; importing `bovnar` never requires it. Install with `pip install "bovnar[pint]"` (or `bovnar[all]`).

The four bridge functions are exported from the top-level `bovnar` namespace (and from `bovnar._pint_bridge`):

Function	Direction	Description
<code>to_pint(value, vu, *, ureg=None)</code>	bovnar → pint	Wrap a scalar/ndarray + <code>ValueUnit</code> in a pint <code>Quantity</code>
<code>to_pint_unit(vu, *, ureg=None)</code>	bovnar → pint	<code>ValueUnit</code> → pint <code>Unit</code> (dimensionless when no real unit)
<code>from_pint(qty, *, ureg=None, validate=True)</code>	pint → bovnar	<code>Quantity</code> → (magnitude, <code>ValueUnit</code>)
<code>from_pint_unit(unit, *, ureg=None, validate=True)</code>	pint → bovnar	<code>Unit</code> / <code>Quantity</code> / <code>str</code> → <code>ValueUnit</code>


```
import bovnar

vu = bovnar.parse_unit("k~m")          # kilometre
qty = bovnar.to_pint(5.0, vu)          # <Quantity(5.0, 'kilometer')>

mag, vu2 = bovnar.from_pint(qty)       # (5.0, ValueUnit for km)
vu3      = bovnar.from_pint_unit("newton") # str/Unit/Quantity all accepted
```

Prefixes ride in the unit, never the magnitude

bovnar's `k~m` maps to pint `kilometer` — the prefix is kept inside the unit name, never folded into the magnitude. A wrapped value is therefore returned unscaled, so a wrapped numpy array is never silently rescaled.

Affine temperature units

Offset/affine scales (`°C`, `°F`, Réaumur, Delisle, Newton, Rømer) cannot carry a prefix or exponent — pint forbids it and bovnar never emits it. A prefixed or exponentiated affine unit raises `BovnarArgumentError` rather than a cryptic pint error.

Validation

`from_pint` / `from_pint_unit` validate the resulting `ValueUnit` by default (`validate=True`); a pint unit that maps to a structurally invalid bovnar unit (e.g. a prefix not permitted on that base) raises `BovnarArgumentError`. pint units with more than 8 components, non-integer exponents, or exponents outside `[-9, 9]` also raise.

Registry control: `build_registry`

A module-level default `pint.UnitRegistry` is built on first use. To share a registry across calls — or to register bovnar's custom units onto your own — pass `ureg=`:

```
from bovnar._pint_units import build_registry, is_currency_unit

ureg = build_registry()          # fresh registry with bovnar units
ureg = build_registry(my_existing_registry) # extend an existing one
ureg = build_registry(with_currencies=False) # skip the currency dimensions

qty = bovnar.to_pint(5.0, vu, ureg=ureg)
```

`build_registry` registers bovnar's custom physical-unit definitions — units where pint's own definition differs or is missing (e.g. `bvnr_gauss`, `bvnr_var`, the historical German units) — plus, by default, the ISO 4217 + crypto currencies as custom dimensions.

`is_currency_unit(unit)` reports whether a pint unit involves a currency dimension (holds for products such as `USD/year`).

Semantic caveats

A few bovnar units map to a pint native unit whose semantics differ slightly; these are recorded in `bovnar._pint_units.SEMANTIC_CAVEATS` (e.g. `BYTE`, `DECIBEL`, `NEPER`). Consult that dict when exact round-trip semantics matter.

The mapping is **not 1:1** by construction (bovnar `b`=bit vs pint barn, `R`=roentgen vs pint's gas constant). The translation table is locked against silent drift by `TestUnitTableIntegrity`, which re-derives every physical unit's dimension and magnitude from bovnar and asserts the pint bridge reproduces both.

NumPy bridge

The NumPy bridge converts between bovnar arrays and `numpy.ndarray`. `numpy` is an **optional** dependency, imported lazily on first use. Install with `pip install "bovnar[numpy]"` (or `bovnar[all]`).

All five functions are exported from the top-level `bovnar` namespace (and from `bovnar._numpy`):

Function	Direction	Description
<code>to_numpy(src, *, dtype=None, return_unit=False)</code>	bovnar → numpy	Array → <code>ndarray</code> (optionally (<code>ndarray, unit_str</code>))
<code>to_pint_array(src, *, dtype=None, ureg=None)</code>	bovnar → pint	Array → pint <code>Quantity</code> (ndarray data + unit)
<code>from_numpy(writer, key, arr, *, unit=None)</code>	numpy → bovnar	Write an <code>ndarray</code> into a <code>Writer</code>
<code>from_pint_array(writer, key, qty)</code>	pint → bovnar	Write a pint <code>Quantity</code> (magnitude + unit) into a <code>Writer</code>
<code>array_to_bvnr(key, arr, *, unit=None, pretty=True)</code>	numpy → bovnar	<code>ndarray</code> → bovnar bytes (convenience)

Reading: `to_numpy`

`src` is either a `DomNode` for an ARRAY (random-access, from `dom_parse`) or the nested list/tuple that `loads(..., typed=True)` produces. Both `/`-separated rows and bracket nesting collapse to the same `ndarray` shape.

```

import bovnar

# from a typed loads() result
doc = bovnar.loads(b'.a=<uint:8>[1,2,3];', typed=True)
arr = bovnar.to_numpy(doc['a'])          # dtype uint8

# with the whole-array unit
arr, unit = bovnar.to_numpy(
    bovnar.loads(b'.a=<float:32,m/s>[1,2,3]/[4,5,6];', typed=True)['a'],
    return_unit=True)                    # unit == 'm/s', arr.shape == (2, 3)

# from a DOM node
arr = bovnar.to_numpy(bovnar.dom_parse(b'.a=<sint:16>[10,-20,30];')['a'])

```

The bovnar (family, width) maps directly to the numpy dtype (uint:8 → uint8, float:32 → float32, ...); FLOAT_FIX / FLOAT_DEC fall back to float64. Pass dtype= to coerce.

A datetime array on the **unix** epoch (spec 1.1) maps to/from datetime64[s] — the integer epoch-seconds carrier is unix-relative, so it round-trips losslessly. A non-unix epoch (tai, gps, ...) is not unix-relative, so to_numpy refuses it (mapping it to datetime64 would mis-date every value) and points you at dtype='int64' for the raw seconds. On the write side a datetime64[*] array is coarsened to whole seconds and written as <datetime:64,unix>; NaT has no bovnar representation and is rejected, and array_to_bvnr prepends the #! bovnar 1.1 directive so its output re-parses.

Strict by default:

- an array that mixes numeric encodings (e.g. uint:8 and float:64) raises unless you pass dtype= to coerce to one;
- bovnar 1.0 arrays are rectangular and homogeneous, so the result is always a regular ndarray — there is no ragged case to handle;
- a null element cannot fill an integer array (dtype=float or dtype=object);
- the unit is a whole-array property (numpy has one dtype per array) — mixed units raise; it is returned alongside the data, never baked into elements.

Writing: from_numpy / array_to_bvnr

```

import numpy as np
import bovnar
from bovnar import Writer

with Writer.to_mem() as w:
    bovnar.from_numpy(w, "velocity",
        np.array([1.5, 2.5], dtype=np.float32), unit="m/s")
out = w.get_output()

# one-shot convenience
raw = bovnar.array_to_bvnr("matrix", np.arange(8).reshape(2, 2, 2))

```

- from_numpy requires a 1-D-or-higher array; write a 0-D scalar with the scalar Writer API instead.

- unit may be a bovnar unit string, a `ValueUnit`, or a pint `Unit` / `Quantity`.
- Units apply to **numeric** arrays only — a unit on a `bool` or string array raises `BovnarArgumentError`.
- `array_to_bvnr` grows its write buffer like `dumps()` (4 MiB, doubling up to 256 MiB).

pint arrays

`to_pint_array` and `from_pint_array` bridge straight through to pint Quantities backed by ndarrays, reusing the unit translation above:

```
q = bovnar.to_pint_array(
    bovnar.loads(b'.a=<float:64,k~m>[1,2,3];', typed=True)['a'])
# q is a pint Quantity: magnitude ndarray [1, 2, 3], unit 'kilometer'

with Writer.to_mem() as w:
    bovnar.from_pint_array(w, "dist", q)
```

Currency helpers

The `bovnar.currency` submodule (exposed as `bovnar.currency`) provides metadata for the ISO 4217 fiat and cryptocurrency `BaseUnit` members. It is pure Python — no library or optional dependency required.

```
from bovnar import currency, BaseUnit

currency.is_currency(BaseUnit.USD) # True
currency.is_fiat(BaseUnit.USD)    # True
currency.is_crypto(BaseUnit.BTC)  # True

info = currency.currency_info(BaseUnit.USD)
info.code          # 'USD'
info.numeric_code  # 840 (ISO 4217 numeric; 0 for crypto)
info.minor_unit    # 2   (decimal places: 1 major = 10^minor minor units)
info.name          # 'US Dollar'

currency.minor_unit(BaseUnit.JPY) # 0
currency.currency_code(BaseUnit.EUR) # 'EUR'
currency.from_code('GBP')          # BaseUnit.GBP

for ci in currency.all_fiat():      # all_crypto() / all_currencies() also exist
    ...
```

`CurrencyInfo` is a frozen dataclass; `currency_info`, `minor_unit`, `currency_name`, `currency_code`, and `from_code` raise for non-currency bases or unknown codes.

Reader reference

Construction

```
with Reader() as r:
    ...
```

`Reader.__init__` calls `bvnr_reader_create` immediately. Use as a context manager or call `r.close()` explicitly to release the C object.

`read_mem(data, *, on_verified, on_unverified, max_file_size, continue_on_error)`

Parse BVNR from a `bytes`, `bytearray`, or `memoryview` object.

Parameter	Type	Default	Description
<code>data</code>	<code>bytes</code> <code>bytearray</code> <code>memoryview</code>	—	Input buffer
<code>on_verified</code>	<code>Callable[[Event, BvnrData] None, bool] None</code>	<code>None</code>	Callback for validated events
<code>on_unverified</code>	<code>Callable[[Event, BvnrData] None, bool] None</code>	<code>None</code>	Callback for raw pre-validation events
<code>max_file_size</code>	<code>int</code>	<code>0</code> (unlimited)	Hard limit on bytes consumed
<code>continue_on_error</code>	<code>bool</code>	<code>False</code>	Enable resync mode

`read_fd(fd, *, on_verified, on_unverified, max_file_size, continue_on_error)`

Parse BVNR from an open POSIX file descriptor. Parameters identical to `read_mem` except the first argument is a non-negative `int` `fd`.

`read_file(path, *, on_verified, on_unverified, max_file_size, continue_on_error)`

Convenience wrapper: opens `path` with `os.O_RDONLY`, calls `read_fd`, closes the `fd` in a `finally` block. The `max_file_size` default is `MAX_FILESIZE_BYTES` (16 MiB) rather than unlimited.

`iter_mem(data, *, verified_only, max_file_size)`

Generator that collects all events from `read_mem` and yields `EventPayload(event, raw, value_type, value_unit)` objects.

Parameter	Default	Description
<code>verified_only</code>	<code>True</code>	When <code>False</code> , both callbacks fire and events may appear twice

Parameter	Default	Description
<code>max_file_size</code>	<code>0</code>	Forwarded to <code>read_mem</code>

`EventPayload` fields:

Field	Type	Description
<code>event</code>	<code>Event</code>	The event code
<code>raw</code>	<code>bytes</code>	Raw token bytes
<code>value_type</code>	<code>ValueTypeSpec \ None</code>	Type annotation if present
<code>value_unit</code>	<code>ValueUnit \ None</code>	Unit if present
<code>text</code>	<code>str</code> (property)	<code>raw</code> decoded as UTF-8 (with <code>errors='replace'</code>)

Error-state properties

These properties query the C reader object after a failed parse.

Property	Type	Description
<code>error_code</code>	<code>ErrorCode</code>	Most recent error code
<code>error_line</code>	<code>int</code>	1-based line number of the error
<code>error_column</code>	<code>int</code>	1-based column of the error
<code>error_offset</code>	<code>int</code>	Byte offset of the error
<code>recovery_count</code>	<code>int</code>	Number of times resync was entered (incremented at error entry, not at resync completion)

MAX_FILESIZE_BYTES

```
from bovnar import MAX_FILESIZE_BYTES  # 16 * 1024 * 1024 (16 MiB)
```

Default `max_file_size` cap used by `Reader.read_file`.

Writer reference

Construction class methods

Method	Description
<code>Writer.to_mem(buf=None, cap=262144, *, pretty=True)</code>	Write to an in-process buffer. <code>buf</code> may be a pre-allocated <code>bytearray</code> ; when <code>None</code> an internal buffer of size <code>cap</code> is allocated.
<code>Writer.to_fd(fd, *, pretty=True)</code>	Write to an open POSIX file descriptor.

Method	Description
<code>Writer.to_file(path, *, pretty=True)</code>	Open <code>path</code> for writing (<code>O_WRONLY\ O_CREAT\ O_TRUNC</code> , mode <code>0o644</code>) and write to it; the fd is closed when the writer is finished or destroyed.

All three are used as context managers. On clean exit (`exc_type is None`) the context manager calls `finish()` automatically.

Output retrieval

Method / property	Description
<code>w.get_output() -> bytes</code>	Return the bytes written so far (mem writers only).
<code>w.bytes_written</code>	Number of bytes written (all writer modes).
<code>w.finish()</code>	Flush and seal the output. Raises <code>BovnarWriteError</code> if any struct is still open.
<code>w.destroy()</code>	Release the underlying C writer object immediately.

Scalar write methods

All scalar writers accept keyword-only unit parameters. Unit resolution priority: `unit_str` (parsed via `bvn_parse_unit_n`) → `unit_iec_base` → `unit_si_base` → `BVN_UNIT_NONE` (no annotation emitted).

```
write_uint(key, value, *, width=64, base=10, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR, unit_iec_base=None,
unit_iec_prefix=IECPrefix.NONE)
```

Write an unsigned integer. `base` selects the numeral system (10 or 16 for inline values; any Bovnar-supported base for `write_bvni`).

```
write_sint(key, value, *, width=64, base=10, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Write a signed integer.

```
write_float(key, value, *, width=64, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Write an IEEE 754 binary float.

```
write_float_fix(key, value, *, width=64, q=0, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Write a Q-format fixed-point value. `q` is the number of fractional bits (`0 ≤ q < width`).

```
write_float_dec(key, value, *, width=64, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Write an IEEE 754-2008 decimal float.

```
write_string(key, value)
```

Write a bare quoted UTF-8 string with no type annotation:

```
.host = "localhost";
```

To emit an explicit `<utf8>` annotation, use the low-level `emit` API with `Event.TYPE_ANNOTATION_START` / `TYPE_ANNOTATION_END` before `Event.DATA`.

```
write_bool(key, value)
```

Write a `bool` value (`token_is_bool`) — serialized as the bare keyword `true` or `false`. On read-back it decodes to a Python `bool`.

```
write_null(key)
```

Write a null value (empty slot).

Extended integer writers

```
write_bvni(key, value, *, width=64, base=10, signed=None, unit_str=None,
unit_si_base=None, unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Arbitrary-width integer writer that supports all Bovnar numeral bases (**2-62, 64, and 85**). Non-decimal values are formatted using Python's own big-integer arithmetic and emitted as quoted strings. `signed` defaults to `True` when `value < 0`.

```
write_bvnf_base(key, value_str, *, width=0, base=10, unit_str=None, unit_si_base=None,
unit_si_prefix=SIPrefix.NONE, unit_si_exp=Exponent.LINEAR)
```

Write a float from a pre-formatted string in base 10 or 16. `base` must be 10 or 16; any other value raises `BovnarArgumentError`.

Struct helpers

```
w.begin_struct(key)    # emit ASSIGNMENT_START + STRUCT_START, increment depth
w.end_struct()         # emit STRUCT_END, decrement depth
```

`finish()` verifies that the struct depth is zero; an unclosed struct raises `BovnarWriteError(GOT_INCOMPLETE_BVNR_STREAM)`.

Array helpers

```
w.begin_array_row()    # emit ARRAY_ROW_START
w.end_array_row()      # emit ARRAY_ROW_END
w.new_array_dim()      # emit ARRAY_DIM_START (the / separator between rows)
```

Low-level `emit`

```
w.emit(event, *, key=None, value=None, vt=None, vu=None)
```

Send an arbitrary event to the C writer. `key` and `value` are encoded as UTF-8. When both `vt` and `value` are supplied the token type is inferred: `_TOKEN_IS_STRING` for `ValueTypeFamily.UTF8`, `_TOKEN_IS_NUMBER` otherwise.

DOM API

The DOM API parses a complete BVNR document into an in-memory tree for random-access queries without writing a SAX callback.

`dom_parse(data) -> DomDoc`

Top-level convenience function (mirrors `DomDoc.parse`).

```
import bovnar
doc = bovnar.dom_parse(bvnr_bytes)
```

DomDoc

Owning wrapper around `bnv_dom_doc_t`. Destroying the object frees the entire tree; any `DomNode` derived from it becomes invalid after that point.

Method / property	Description
<code>DomDoc.parse(data)</code>	Class method. Parse <code>bytes</code> <code> </code> <code>bytearray</code> <code> </code> <code>memoryview</code> .
<code>DomDoc.parse_fd(fd)</code>	Class method. Parse from an open file descriptor.

File-size limits: `DomDoc.parse_fd` and `DomDoc.parse_file` apply an internal ceiling of `BNV_DOM_FD_MAX_BYTES` (256 MiB). This is 16× larger than `Reader.read_file`'s `MAX_FILESIZE_BYTES` (16 MiB). Use `bnv_dom_parse_fd_ex` directly if you need a different limit. `| DomDoc.parse_file(path) |` Class method. Open path and parse (fd closed in `finally`). Applies an internal hard cap of `BNV_DOM_FD_MAX_BYTES` (256 MiB) via the C function `bnv_dom_parse_fd`. This ceiling is distinct from `Reader.read_file`'s `MAX_FILESIZE_BYTES` (16 MiB). `| | doc.parse_error | ErrorCode — NONE on success. | | doc[key] |` Return top-level `DomNode` by key; raises `KeyError` when absent. `| | key in`

`doc` | `True` when the top-level key exists. | `len(doc)` | Number of top-level entries. | `iter(doc)` | Iterate over `(key, DomNode)` pairs. | `doc.entries()` | Return all top-level `(key, DomNode)` pairs as a list. | `doc.lookup(path)` | Dot-separated path lookup, e.g. `'server.tls.cert'`. Returns `None` when absent. | `doc.to_dict()` | Convert entire document to a plain Python dict, dropping type and unit info. |

DomNode

Non-owning view into a `bvn_dom_node_t`. The parent `DomDoc` must remain alive for as long as any derived `DomNode` is in use.

Property / method	Description
<code>node.dom_type</code>	<code>DomType</code> enum value
<code>node.value_type</code>	<code>ValueTypeSpec</code>
<code>node.unit</code>	<code>ValueUnit</code>
<code>node.unit_str</code>	Unit as a string via <code>bvn_dom_get_unit_string</code> , or <code>''</code>
<code>node.is_null()</code>	<code>True</code> for null values
<code>node.value_in_base_units()</code>	Numeric value scaled to SI base units (<code>float</code>)
<code>node.as_i64()</code> / <code>as_u64()</code>	Signed / unsigned 64-bit integer
<code>node.as_i32()</code> / <code>as_u32()</code>	Signed / unsigned 32-bit integer
<code>node.as_i16()</code> / <code>as_u16()</code>	Signed / unsigned 16-bit integer
<code>node.as_i8()</code> / <code>as_u8()</code>	Signed / unsigned 8-bit integer
<code>node.as_float()</code>	<code>float</code> (64-bit)
<code>node.as_str()</code>	Python <code>str</code> for <code>STRING</code> , <code>SYMBOL</code> , or <code>REFERENCE</code> nodes
<code>node.as_bytes()</code>	<code>bytes</code> for <code>OCTET_STREAM</code> nodes
<code>node.as_int_str(base=10)</code>	Integer value as a string in the given base; result C string is freed before return
<code>node.datetime_fraction</code>	<code>str</code> of the verbatim ISO sub-second digits for a <code>datetime</code> written as a literal with a fraction (spec 1.1), else <code>None</code> ; the carrier value is unchanged and still read via <code>to_python()</code>
<code>node[key]</code>	Child <code>DomNode</code> by string key (STRUCT) or integer index (ARRAY)
<code>key in node</code>	Membership test for STRUCT nodes
<code>len(node)</code>	Element count for STRUCT or ARRAY nodes
<code>iter(node)</code>	For STRUCT: iterate <code>(key, DomNode)</code> pairs; for ARRAY: iterate elements
<code>node.entries()</code>	List of <code>(key, DomNode)</code> pairs (STRUCT nodes only)
<code>node.array_dims()</code>	Number of <code>/</code> -separated dimensions (ARRAY nodes only)

Property / method	Description
<code>node.to_python()</code>	Recursively convert to a native Python value (drops type/unit info)

DomType

```

NULL=0  INT=1  FLOAT=2  STRING=3  SYMBOL=4
REFERENCE=5  STRUCT=6  ARRAY=7  OCTET_STREAM=8  BOOL=9

```

Running the test suite

```

# Run everything (library-dependent tests are skipped when libbvnr_shared.so is absent)
pytest

# Run only the pure-Python tests (no library required)
pytest -m "not needs_lib"

# Run only integration tests (requires libbvnr_shared.so)
pytest -m needs_lib

# Verbose with short tracebacks (already the default via pyproject.toml)
pytest -v --tb=short

```

The NumPy- and pint-bridge tests `pytest.importorskip` their dependency, so they are skipped automatically when `numpy` / `pint` is not installed. Install `bovnar[all]` to run the full suite.

Package layout

```

bovnar/
├── __init__.py      # loads() / dumps() / dom_parse() / unit helpers – public API
├── ffi.py           # ctypes FFI: library discovery + argtypes/restype
├── numpy.py         # NumPy bridge: to_numpy/from_numpy/to_pint_array/array_to_bvnr
(numpy optional)
├── pint_bridge.py   # pint bridge: to_pint/from_pint/to_pint_unit/from_pint_unit (pint
optional)
├── pint_units.py    # verified bovnar↔pint unit table + build_registry()
├── currency.py      # ISO 4217 / crypto currency metadata (pure Python)
├── dom.py           # DomDoc, DomNode, DomType – random-access DOM
├── enums.py         # Python IntEnum mirrors of C enums
├── exceptions.py    # BovnarError hierarchy
├── quantity.py      # Quantity – typed, unit-annotated scalar for lossless round-trips
├── reader.py        # Reader class + EventPayload dataclass
├── structs.py       # ctypes Structure/Union definitions + ValueUnitPrefix + make_*
helpers
├── units.py         # unit_to_si_factor, unit_convert_factor, etc.
├── writer.py        # Writer class

tests/
├── conftest.py      # shared fixtures, needs_lib marker
├── test_analytics.py # analytic / benchmarking tests (needs_lib)
├── test_array_parser.py # array parsing integration tests (needs_lib)
├── test_currency_units.py # currency unit round-trip tests (needs_lib)
├── test_dom.py       # DOM API integration tests (needs_lib)
├── test_enums.py     # pure-Python enum tests
├── test_example_numpy_roundtrip.py # NumPy bridge end-to-end example (needs_lib + numpy)
├── test_numpy_bridge.py # NumPy bridge tests (needs_lib + numpy)
├── test_pint_bridge.py # pint bridge + unit-table integrity (needs_lib + pint)
├── test_reader.py    # integration: Reader (needs_lib)
├── test_structs.py   # pure-Python struct / helper tests
├── test_unit_physics.py # unit physics / conversion tests (needs_lib)
├── test_units.py     # mixed: unit parsing / serialisation (needs_lib for
FFI)
├── test_write_array.py # write_array integration tests (needs_lib)
├── test_writer.py    # integration: Writer (needs_lib)

```

Error handling

All errors surface as subclasses of `BovnarError`:

Exception	When raised
<code>BovnarLibraryNotFound</code>	<code>libbvnr_shared.so</code> not found at import
<code>BovnarParseError</code>	Parse error in <code>Reader</code> (carries <code>code</code> , <code>line</code> , <code>column</code> , <code>offset</code> , <code>byte</code>)
<code>BovnarWriteError</code>	Write error in <code>Writer</code> (carries <code>code</code> , <code>offset</code>)
<code>BovnarArgumentError</code>	Invalid argument passed to a helper (e.g. bad unit string, closed reader/writer)

`unit_convert_factor` error semantics: The C function uses the `(ok, requires_affine)` pair to signal three distinct outcomes:

ok	requires_affine	Meaning
True	False	Multiplicative conversion; <code>factor</code> is ready to use
False	True	Affine conversion required (e.g. °C ↔ K); <code>factor</code> is still valid but a plain multiply is insufficient
False	False	Units are dimensionally incompatible → <code>BovnarArgumentError</code>

`BovnarArgumentError` is raised **only** when both `ok=False` and `requires_affine=False`. When `requires_affine=True`, call `convert_value` which handles the two-step affine path automatically, or call `unit_to_si_factor` on both units and perform the conversion manually.

Callbacks returning `False`: When an `on_verified` or `on_unverified` callback returns `False`, the C parser stops and `bvnr_read` returns failure. The Python layer then calls `_raise_error()` and raises `BovnarParseError` with whatever error code the reader recorded. `BovnarCallbackAbort` is defined in `exceptions.py` but is not raised by the current implementation.

Callbacks raising exceptions: If the callback itself raises a Python exception, that exception is captured, the C callback returns `False` to stop the parse, and the original exception is re-raised from `read_mem` / `read_fd` after the C call returns.

FFI details

`ON_ERROR_FUNC` signature

The error callback type matches the C `bvnr_on_error_fn` signature exactly:

```
void (*bvnr_on_error_fn)(
    void*      userdata,
    error_code_t err,
    uint64_t   line,
    uint64_t   column,
    uint32_t   byte,
    uint64_t   offset);
```

In Python this is declared as:

```
ON_ERROR_FUNC = ctypes.CFUNCTYPE(
    None,
    ctypes.c_void_p,    # userdata
    ctypes.c_int,       # error_code_t err
    ctypes.c_uint64,    # line
    ctypes.c_uint64,    # column
    ctypes.c_uint32,    # byte
    ctypes.c_uint64,    # offset
)
```

`BvnrWriteFlags` layout

`BvnrWriteFlags` mirrors the C `bvnr_write_flags_s` struct in full, including the trailing `unit_flags` field (`bvn_unit_flags_t`, a `uint32_t`):

Flag constant (C)	Value	Effect
<code>BVN_UNIT_FLAGS_NONE</code>	0	Default: full Unicode exponent characters
<code>BVN_UNIT_REDUCE</code>	<code>1 << 0</code>	Reduce compound units before serialising
<code>BVN_UNIT_ASCII_EXP</code>	<code>1 << 1</code>	Use <code>^N</code> ASCII exponent notation instead of Unicode superscripts

The writer respects the `unit_flags` stored inside the C writer object when serialising unit annotations. `_emit_annotation` retrieves the live flags via `bvnr_writer_unit_flags(w)` and passes them to `bvn_unit_to_string_ex`.

`write_string` behaviour

`Writer.write_string` emits a bare quoted string with no type annotation, matching `bvnr_write_string` in the C library:

```
.host = "localhost";
```

To write a string with an explicit `<utf8>` annotation, use the low-level `emit` API with `Event.TYPE_ANNOTATION_START` / `TYPE_ANNOTATION_END` events before the `Event.DATA` event.

`_resolve_unit` default

When no unit arguments are supplied to `write_uint`, `write_sint`, `write_float`, etc., the unit resolves to `BVN_UNIT_NONE` (zero components), matching the C convenience helpers. No unit annotation is emitted in this case. Passing `unit_si_base` or `unit_iec_base` produces a unit with one component; the `_SENTINEL`-only `no_unit` keyword is only produced when the caller explicitly constructs a dimensionless `ValueUnit` with `BaseUnit.NONE`.

`BaseUnit` enum

The `BaseUnit` enum mirrors the full C `value_base_unit_e`:

Range	Members
0	<code>NONE</code>
1-2	<code>BIT</code> , <code>BYTE</code>
3-9	SI base units (<code>SECOND</code> ... <code>CANDELA</code>)

Range	Members
10–28	Named SI derived units (<code>HERTZ</code> ... <code>KATAL</code>)
29–44	Non-SI accepted units (<code>LITER</code> ... <code>YEAR</code>)
45–54	Imperial/US length (<code>INCH</code> ... <code>FATHOM</code>)
55–62	Imperial/US mass (<code>POUND</code> ... <code>CARAT</code>)
63	<code>FAHRENHEIT</code>
64–67	Pressure (<code>ATMOSPHERE</code> ... <code>PSI</code>)
68–71	Energy (<code>CALORIE</code> ... <code>THERM</code>)
72	<code>HORSEPOWER</code>
73–75	Force (<code>POUND_FORCE</code> , <code>DYNE</code> , <code>KIP</code>)
76	<code>KNOT</code>
77–85	US volume (<code>GALLON</code> ... <code>BARREL</code>)
86–87	Area (<code>ACRE</code> , <code>BARN</code>)
88–90	Angle (<code>ARCMINUTE</code> , <code>ARCSECOND</code> , <code>GRAD</code>)
91–98	CGS units (<code>POISE</code> ... <code>GALILEO</code>)
99–101	Radiation (<code>CURIE</code> , <code>ROENTGEN</code> , <code>REM</code>)
102–103	Logarithmic (<code>NEPER</code> , <code>DECIBEL</code>)
104	<code>RANKINE</code>
105	<code>SLUG</code>
106	<code>THOU</code>
107–109	UK imperial volume (<code>PINT_UK</code> , <code>FLUID_OUNCE_UK</code> , <code>QUART_UK</code>)
110–111	Electrical power (<code>VAR</code> , <code>VOLT_AMPERE</code>)
112	<code>KILOGRAM_FORCE</code>
113	<code>INCH_HG</code>
114	<code>RPM</code>
115	<code>FOOT_POUND</code>
116–117	Mass additional (<code>DRAM</code> , <code>PENNYWEIGHT</code>)
118–119	Length additional (<code>CHAIN</code> , <code>ROD</code>)
120–121	Volume additional (<code>GILL</code> , <code>GILL_UK</code>)
122	<code>STANDARD_GRAVITY</code>
123	<code>METRIC_HORSEPOWER</code>
124	<code>REVOLUTION</code>
125–126	Time additional (<code>MONTH</code> , <code>FORTNIGHT</code>)

Range	Members
127	<code>ATMOSPHERE_TECHNICAL</code>
128–129	Textile linear density (<code>TEX</code> , <code>DENIER</code>)
130–133	Apothecary/dry volume (<code>FLUID_DRAM</code> , <code>MINIM</code> , <code>PECK</code> , <code>BUSHEL</code>)
134–297	ISO 4217 fiat currencies (<code>AED</code> ... <code>ZWL</code>) — see <code>CURRENCY_FIAT_FIRST</code> / <code>CURRENCY_FIAT_LAST</code>
298–347	Cryptocurrencies (<code>BTC</code> ... <code>RUNE</code>) — see <code>CURRENCY_CRYPTO_FIRST</code> / <code>CURRENCY_CRYPTO_LAST</code>
348–360	Historical German units (<code>PFUND</code> , <code>ZENTNER</code> , <code>LOT</code> , Prussian line/zoll/fuss/elle/rute, <code>KLAFTER</code> , <code>GERMAN_MILE</code> , <code>MORGEN</code> , <code>SCHEFFEL</code>)
361–367	Additional physical units (<code>SURVEY_FOOT</code> , <code>LEAGUE</code> , <code>CABLE</code> , <code>HAND</code> , <code>QUINTAL</code> , <code>SCRUPLE</code> , <code>BAUD</code>)
368–371	Temperature scales (<code>DELISLE</code> , <code>NEWTON_TEMP</code> , <code>REAUMUR</code> , <code>ROMER</code>)
372–377	Ratio/proportion units (<code>PERCENT</code> , <code>PER_MILLE</code> , <code>PER_MYRIAD</code> , <code>PER_CENT_MILLE</code> , <code>PPM</code> , <code>PPB</code>)
378	<code>_SENTINEL</code> (internal bound; do not use)

Note on `CUP`: the Cuban-Peso currency is exposed as `BaseUnit.CUP_` (trailing underscore), not `BaseUnit.CUP`. The plain name `CUP` is the US-cup volume unit (enum value 81); since Python enum member names must be unique, the currency at value 167 takes the suffixed name. This affects the Python member name only — the `.bvnr` wire token is still the bare uppercase `CUP`, and case stays load-bearing (`cup` = volume, `CUP` = currency).

Spec 1.1 additions

These bindings target the **Bovnar spec (v1.1, draft)**; spec 1.0 remains the frozen baseline. The 1.1 features (all gated on a `#!bovnar 1.1` directive — an unversioned document is treated as 1.0) are exposed as:

- **Version:** `bovnar.version()` (library version string), `bovnar.spec_version()` → (major, minor) of the highest spec understood, and `bovnar.peek_version(data)` → the (major, minor) a document declares, or `None`. `Reader.declared_version` gives the same after a read; `read_mem` / `read_fd` / `loads` accept `strict_version=True` to reject a too-new version (`ErrorCode.UNSUPPORTED_SPEC_VERSION`).
- **Richer escapes:** `\u{...}` and `\xHH` in string literals are decoded by the C reader, so `loads` returns the resulting text transparently. A surrogate / out-of-range `\u` is `ErrorCode.INVALID_CODEPOINT`.

- **Datetime family:** `loads` decodes a `<datetime:width,epoch>` value to a plain `int` (signed epoch seconds). With `typed=True` it is a `Quantity` whose `.epoch_name` (`"unix"`, `"tai"`, ...) and `.epoch_mjd` recover the epoch. `dumps()` emits the `<datetime:...>` annotation and prepends `#!bovnr 1.1` automatically when the object contains a datetime, so the **carrier** round-trips losslessly. Note the high-level value is only the whole-second carrier: the sub-second fraction of an ISO-8601 literal (spec 1.1) is not part of the `int / Quantity` value, so `loads` → `dumps` drops it. To read or round-trip the fraction use the DOM tier (`DomNode.datetime_fraction`) or the streaming reader (`bvnr_data_t.frac_data` via a callback) — the C pretty-printer preserves it because it re-serialises the event stream, not a materialised value. `ValueTypeFamily.DATETIME` is the family enum member.
- **Reference array indexing:** `&.matrix[0][1]` paths are stored verbatim and resolved by `DomDoc.lookup("matrix[0][1]")` at the DOM layer.

`ErrorCode` adds `INVALID_SPEC_VERSION` (42), `UNSUPPORTED_SPEC_VERSION` (43), `INVALID_CODEPOINT` (44), `INVALID_DATETIME_LITERAL` (45), and `DATETIME_LITERAL_UNSUPPORTED_EPOCH` (46).